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Unit 2 - Directional Control: Clockwise and Counterclockwise

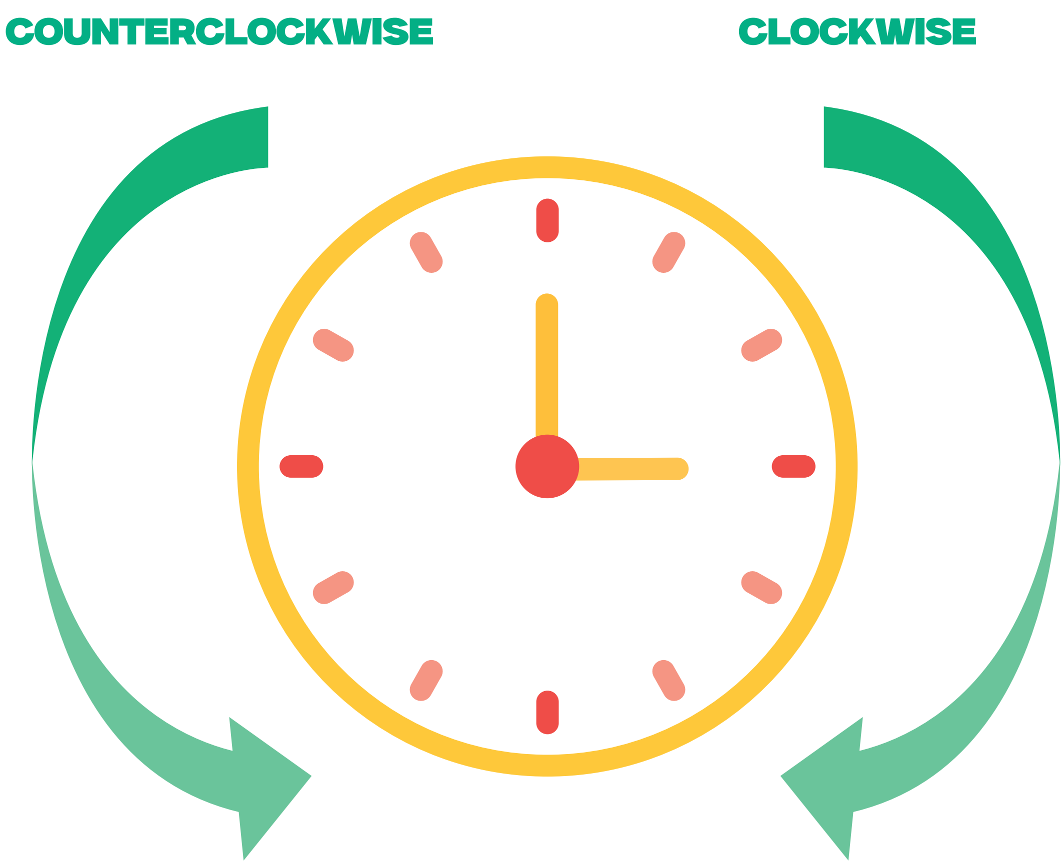
PDF Documents

PDF Document 1: 1738217041876-Chapter 2 - Unit 2 - Directional Control Clockwise and Counterclockwise.pdf

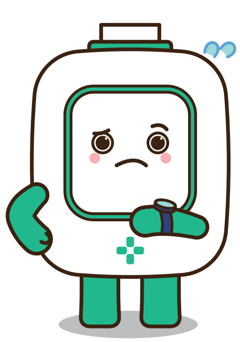
This PDF document is included in the unit materials.

UNIT 2 DIRECTIONAL CONTROL: CLOCKWISE AND COUNTERCLOCKWISE

Understanding Directional Movements



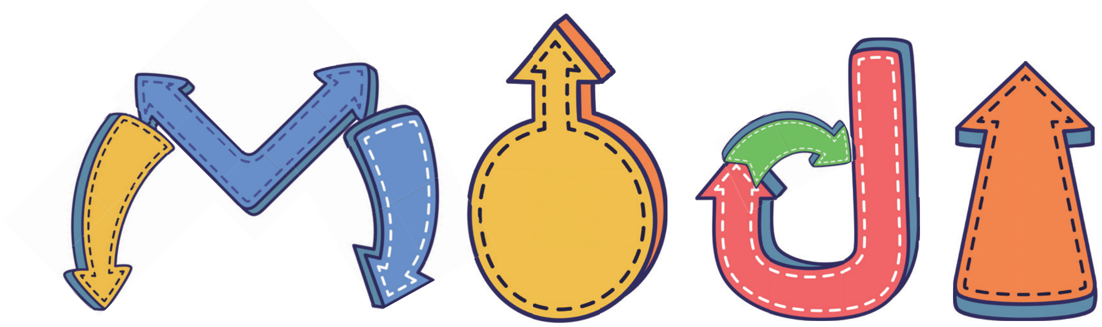
Clockwise movement refers to turning in the same direction as the hands of a clock, while counterclockwise is the opposite direction. Windshield wipers use both of these movements to clean the entire windshield effectively. By moving clockwise and counterclockwise, the wipers can sweep over the entire surface of the glass, pushing water and debris from the center to the edges. This dual-direction movement ensures that all areas of the windshield are covered, leaving the driver with a clear and unobstructed view, which is crucial for safe driving.



MECHANISMS BEHIND DIRECTIONAL CONTROL

Windshield wipers are driven by a setup that includes a motor, linkage, and gears. The motor powers the movement, and the gears translate the motor's rotational energy into the mechanical force that moves the wipers back and forth across the windshield. The linkage, a system of arms and pivots, connects the motor and gears to the wiper blades, facilitating their sweeping motion.

The direction of the motor's rotation can be reversed to change the movement of the blades. This reversal allows the wipers to move both clockwise and counterclockwise, enabling them to cover the entire windshield efficiently. By controlling the direction of the motor's rotation, the system ensures that the wipers can adapt their movement to remove debris and water most effectively, maintaining clear visibility through the windshield.



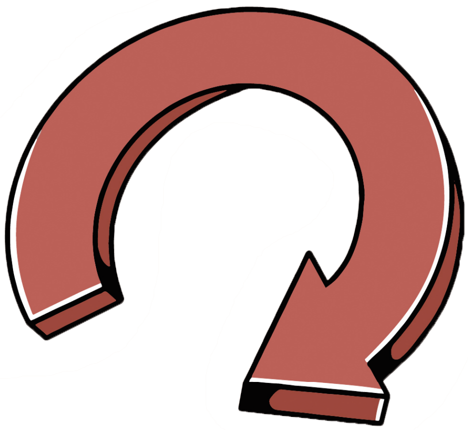
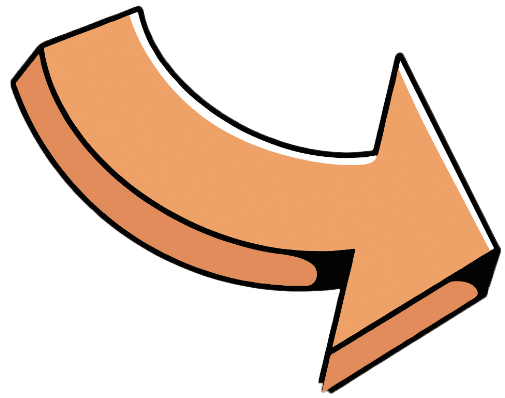
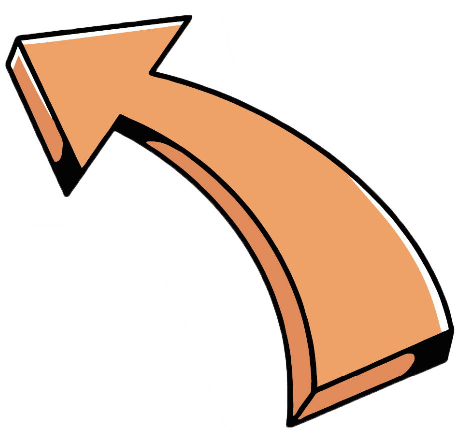
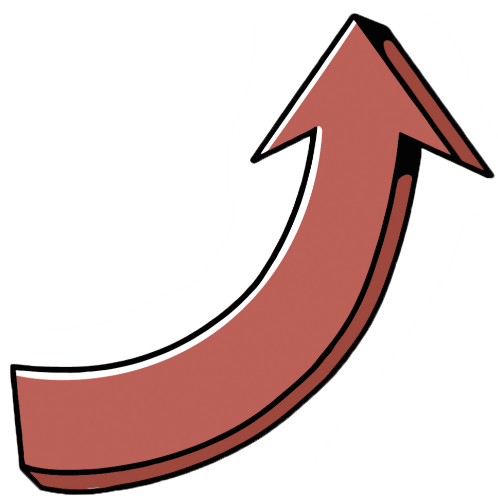
IMPORTANCE OF DIRECTIONAL MOVEMENTS IN CLEANING

Different directional movements of windshield wipers enhance their efficiency in cleaning the glass surface. By moving both clockwise and counterclockwise, the wipers can sweep across the entire windshield, covering all areas. This dual-direction motion ensures that water and debris are pushed efficiently from the center to the edges, minimizing streaks and gaps.

The ability to switch directions helps the wipers maintain consistent contact with the windshield's contour and prevents spots from being missed, resulting in a clearer and more thoroughly cleaned windshield. This comprehensive coverage is crucial for maintaining optimal visibility and safety, especially during adverse weather conditions.

Activity:

Indicate whether the arrows in the table are clockwise or counterclockwise.

Activity: Let's create a wiper!

Materials: Paper plate Markers Scissors

INSTRUCTIONS:

1. **DRAW A WINDSHIELD** Color the center of your paper plate blue to make it look like a car window.

2. **DRAW A WIPER ARM** Draw a long line from the edge of the plate to the center, like a wiper arm.

3. **SHOW THE MOVEMENT**

   Clockwise: With your marker, draw a big circle around the center of the plate, going clockwise. This is how a wiper moves to clean the right side of the windshield.

   Counterclockwise: Now, draw another big circle, but this time going counterclockwise. This is how the wiper cleans the left side of the windshield.